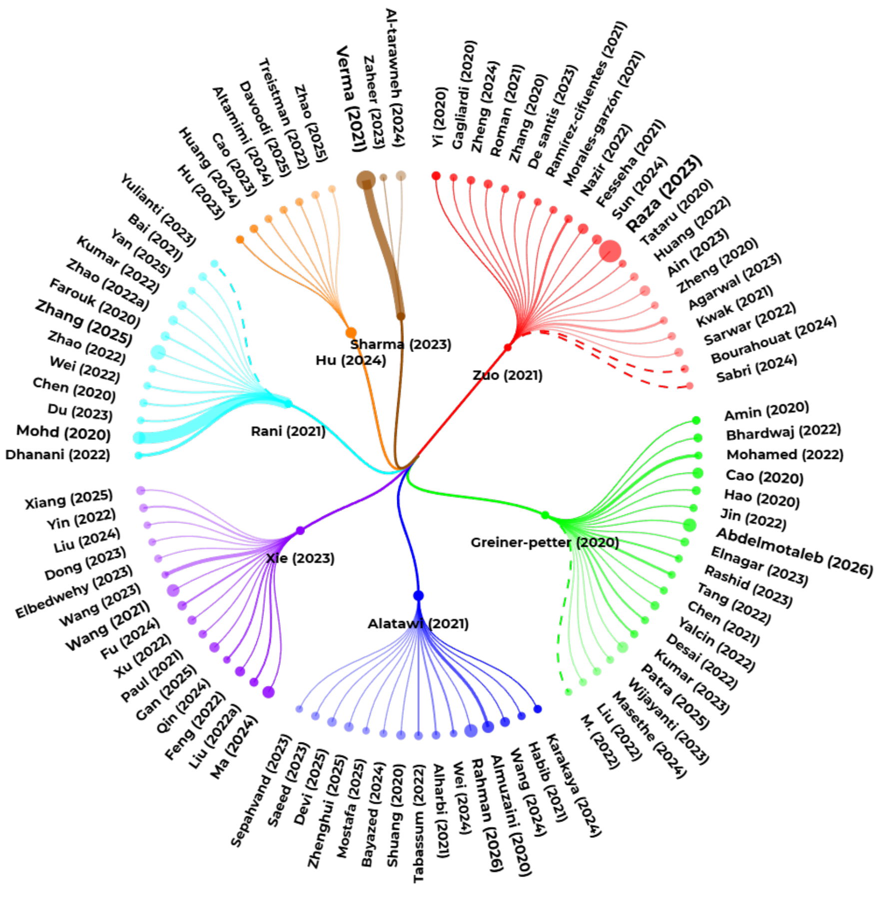
Literature Review and research themes

Your query

[TITLE=("embedding*" AND ("word embedding" OR "semantic embedding")) AND YEAR=(2016-2026) AND RANKING=(2* OR 3* OR 4*)]

Star Map



Number of documents: 106 (threshold: 1.0229276895943562) | Year range: 2020 - 2026

Theme 1 / Versatile Applications of Word Embedding in NLP

Summary:

This theme comprises a diverse set of citations primarily focused on the application of word embedding techniques in various natural language processing (NLP) tasks. These tasks include text classification, topic modeling, keyphrase extraction, sentiment analysis, and more.

Synthesis of all articles in this group:

The cluster of citations on Versatile Applications of Word Embedding in NLP highlights the diverse applications of word embedding techniques in various natural language processing (NLP) tasks. According to Fesseha (2021), word embedding techniques such as Word2Vec can improve text classification accuracy, achieving an accuracy of 93.41% in Tigrinya news classification. Similarly, Roman (2021) found that BERT embedding performed significantly better than other word embedding methods in text classification tasks. The use of word embedding has also been explored in topic modeling and analysis, with Yi (2020) proposing methods to improve topic modeling for short texts by leveraging pre-trained word embeddings. Additionally, word embedding approaches have been applied in sentiment analysis and opinion mining to detect substance abuse and mental health issues in financial texts and social media writings (Zheng, 2024; Ramírez-Cifuentes, 2021). Keyphrase extraction algorithms have also been improved by integrating word embedding with clustering methods (Gagliardi, 2020; Sarwar, 2022). Furthermore, word embedding has been used in language and knowledge graph construction, with De Santis (2023) proposing the use of granular computing and conceptual space theory to improve text categorization, and Agarwal (2023) using a genetic algorithm-based service clustering methodology that assimilates dirichlet multinomial mixture model with word embedding. Morales-Garzón (2021) also demonstrated the potential of specific-domain semantic models using word embedding models to adapt cooking recipes to user preferences. Finally, word embedding techniques have been applied in biological and health applications, such as improving inflammatory bowel disease analysis (Tataru, 2020). Overall, these studies demonstrate the effectiveness and versatility of word embedding techniques in various NLP tasks, highlighting their potential to improve the accuracy and effectiveness of NLP models. The applications of word embedding are diverse and continue to expand, with new studies exploring its use in various domains, including text classification, topic modeling, sentiment analysis, keyphrase extraction, language and knowledge graph construction, and biological and health applications.

Subthemes:

• ***Text Classification :*** (Raza A. (2023) [11], Ain Q. (2023) [12], Fesseha A. (2021) [13], Sun G. (2024) [14], Roman M. (2021) [15], Sabri T. (2024) [9], Bourahouat G. (2024) [3])
The subtheme of Text Classification has been explored in various studies, including the use of word embedding techniques such as Word2Vec (Fesseha, 2021) and BERT (Roman, 2021) to improve text classification accuracy. For instance, Fesseha (2021) achieved an accuracy of 93.41% using a CNN with a continuous bag-of-words method and word2vec for Tigrinya news classification. Similarly, Roman (2021) found that BERT embedding performed significantly better than other word embedding methods, with a precision score of 89%. Other studies have also demonstrated the effectiveness of word embedding in text classification, including the use of TF-IDF and Word2Vec

embedding technique for Arabic text classification (Sabri, 2024), and the application of word embedding as a semantic feature extraction technique in Arabic Natural Language Processing (Bourahouat, 2024). Additionally, the use of hybrid word embedding techniques, such as fastText and Transformer Encoder-Based Hybrid Word Embedding, has been shown to improve the accuracy of text classification models (Raza, 2023). Overall, these studies highlight the importance of word embedding in text classification and demonstrate its potential to improve the accuracy and effectiveness of text classification models. The original title of the subtheme is: Text Classification

• ***Topic Modeling and Analysis :*** (Yi F. (2020) [17], Huang L. (2022) [18], Zhang P. (2020) [23], Zuo Y. (2021) [29])
Topic modeling is a widely studied problem in text mining domains (Yi, 2020). Conventional topic models are limited by data sparsity in short texts, which can be overcome by leveraging pre-trained word embeddings (Yi, 2020). Researchers have proposed various methods to improve topic modeling for short texts, including the use of word embedding and document correlation (Yi, 2020), piecewise linear representation and word embedding (Huang, 2022), and combining topic modeling with semantic embedding (Zhang, 2020). These methods have shown promising results in terms of topic coherence and text classification tasks (Yi, 2020). Additionally, pseudo-document views with word embedding enhancement have been explored for topic modeling of short texts (Zuo, 2021). Overall, these studies demonstrate the effectiveness of incorporating word embeddings and other techniques to improve topic modeling for short texts.

• ***Sentiment Analysis and Opinion Mining :*** (Zheng J. (2024) [19], Ramírez-Cifuentes D. (2021) [30])
Sentiment analysis is applied in financial texts (Zheng, 2024) and social media writings (Ramírez-Cifuentes, 2021) to detect substance abuse and mental health issues. Word embedding approaches are used to enhance detection accuracy. Original title: Sentiment Analysis and Opinion Mining

• ***Keyphrase Extraction and Semantic Analysis :*** (Sarwar T. (2022) [16], Gagliardi I. (2020) [28])
Keyphrase extraction algorithms extract relevant information from documents (Sarwar, 2022). Integrating word embedding with clustering methods improves unsupervised keyphrase extraction (Gagliardi, 2020).

• ***Language and Knowledge Graph Construction :*** (De Santis E. (2023) [22], Agarwal N. (2023) [24], Morales-Garzón A. (2021) [25])
According to De Santis (2023), text categorization can be improved by using granular computing and conceptual space theory. Agarwal (2023) proposes a genetic algorithm-based service clustering methodology that assimilates dirichlet multinomial mixture model with word embedding. Morales-Garzón (2021) uses word embedding models to adapt cooking recipes to user preferences, demonstrating the potential of specific-domain semantic models. These studies highlight the importance of word embedding in various applications, including text categorization and recipe adaptation. The original title of the subtheme is: Language and Knowledge Graph Construction

• ***Biological and Health Applications :*** (Tataru C. (2020) [26])
Tataru (2020) applies word-embedding techniques to microbiomes, improving inflammatory bowel disease analysis.

Articles that do not belong to any subtheme:

• ***versatile applications of word embedding in nlp :*** (Zheng C. (2020) [27])
No summary available.

Theme 2 / Development and Application of Word Embedding Techniques

Summary:

This theme comprises a diverse set of citations primarily focused on the application and development of word embedding techniques across various domains. These include but are not limited to text classification, sentiment analysis, plagiarism detection, health informatics, and semantic search.

Synthesis of all articles in this group:

The cluster of citations on Development and Application of Word Embedding Techniques explores various applications and techniques of word embedding in natural language processing tasks. According to Abdelmotaleb (2026), word embedding techniques can achieve varying levels of accuracy in text classification tasks. Patra (2025) and M. S. (2022) demonstrate the effectiveness of word embedding techniques in phishing email detection and classification, respectively. Yalcin (2022) and Greiner-Petter (2020) show the promise of word embedding in plagiarism detection and semantic search tasks. Desai (2022) highlights the potential of AI models in improving stroke diagnosis. The citations also discuss various word embedding models and techniques, including contextualized word embedding models (Elnagar, 2023), context-aware embedding techniques (Masethe, 2024), and bilingual word embedding methods (Wijayanti, 2023). Furthermore, Liu (2022), Tang (2022), Chen (2021), and Jin (2022) demonstrate the effectiveness of semantic embeddings in improving text retrieval and alignment tasks. Overall, the studies in this cluster demonstrate the effectiveness of word embedding techniques in various natural language processing tasks, including text classification, sentiment analysis, plagiarism detection, semantic search, and information retrieval. However, as noted by Masethe (2024) and Wijayanti (2023), there is still a need for further research to improve the performance of these techniques and address the challenges of low-resource languages.

Subthemes:

• ***Text Classification and Sentiment Analysis :*** (Abdelmotaleb H. (2026) [31], Patra C. (2025) [32], Amin S. (2020) [38], Hao Y. (2020) [35], M. S. (2022) [10])
The subtheme of Text Classification and Sentiment Analysis is explored in various studies. Abdelmotaleb (2026) compared word embedding techniques for classification of star ratings, finding that different techniques can achieve varying levels of accuracy. Patra (2025) used transformer-based word embedding for phishing email detection, leveraging vector similarity search. Amin (2020) proposed a novel approach for disease detection based on social media posts using LSTM and word embedding techniques, achieving 94% accuracy. Hao (2020) introduced stochastic word embedding for cross-domain sentiment encoding. M. S. (2022) used word embedding and machine learning techniques for phishing email classification, achieving an accuracy of 99.50% with a false positive rate of 0.053%. These studies demonstrate the effectiveness of word embedding techniques in text classification and sentiment analysis tasks. The original title of the subtheme is: Text Classification and Sentiment Analysis

• ***Plagiarism Detection and Semantic Search :*** (Yalcin K. (2022) [34], Greiner-Petter A. (2020) [40], Mohamed E. (2022) [44])
Plagiarism detection and semantic search are crucial tasks in natural language processing. Yalcin (2022) proposed a plagiarism detection system based on

part-of-speech (POS) tag n-grams and word embedding. Greiner-Petter (2020) explored math-word embedding in math search and semantic extraction, showing promise for similarity and search tasks. Mohamed (2022) developed a Quranic semantic search tool based on word embedding, achieving high precision and recall. These studies demonstrate the effectiveness of word embedding in various applications, including plagiarism detection and semantic search.

• ***Health Informatics :*** (Desai A. (2022) [47])
AI model improves stroke diagnosis (Desai, 2022)

• ***Word Embedding Models and Techniques :*** (Elnagar A. (2023) [37], Masethe M. (2024) [39], Kumar Y. (2023) [41], Rashid J. (2023) [46], Wijayanti R. (2023) [48])
The citations discuss various word embedding models and techniques, including contextualized word embedding models for Arabic (Elnagar, 2023), context-aware embedding techniques for addressing Meaning Conflation Deficiency in morphologically rich languages (Masethe, 2024), stacking ensemble-based HIDS framework for detecting anomalous system processes using multiple word embedding (Kumar, 2023), word embedding-based topic model with modified collapsed Gibbs sampling for short text (Rashid, 2023), and learning bilingual word embedding for automatic text summarization in low resource language (Wijayanti, 2023). According to Masethe (2024), context-aware embeddings have promise in treating Meaning Conflation Deficiency, but there is a great deal of variability and uncertainty in the available data. Elnagar (2023) presents a benchmark for evaluating Arabic contextualized word embedding models, while Rashid (2023) proposes a novel word embedding-based topic model for short text documents. Kumar (2023) uses multiple word embedding for detecting anomalous system processes, and Wijayanti (2023) investigates bilingual word embedding methods for Indonesian-English language. The results show that these techniques can be effective in various natural language processing tasks, but there is still a need for further research to improve their performance and address the challenges of low-resource languages (Wijayanti, 2023).

• ***Information Retrieval and Text Summarization :*** (Liu L. (2022) [43], Tang X. (2022) [49], Chen J. (2021) [45], Jin Q. (2022) [42])
Recent studies have focused on improving text alignment and retrieval using semantic embeddings. Liu (2022) proposed Bertalign, a two-step algorithm for bilingual sentence alignment, which achieved the highest accuracy in terms of F1 score. Tang (2022) developed a short text matching model with multiway semantic interaction based on multi-granularity semantic embedding. Chen (2021) augmented ontology alignment by semantic embedding and distant supervision. Jin (2022) explored funding patterns with word embedding-enhanced organization–topic networks. These studies demonstrate the effectiveness of semantic embeddings in improving text retrieval and alignment tasks.

Theme 3 / Word Embedding in Deep Learning for NLP and Interdisciplinary Applications

Summary:

This theme comprises a collection of citations primarily focused on the application of word embedding techniques in various domains, including but not limited to natural language processing, bioinformatics, software development, and medical applications.

Synthesis of all articles in this group:

The cluster 'Word Embedding in Deep Learning for NLP and Interdisciplinary Applications' encompasses a broad range of applications, including Natural Language Processing (NLP), software development, bioinformatics, time series classification, and medical and health applications. According to Rahman (2026), word embedding feature extraction can be utilized to identify blood-brain barrier penetrating peptides. Almuzaini (2020) found that stem-based algorithms perform slightly better than word embedding in deep learning-based Arabic text categorization. The use of word embedding in NLP applications has been demonstrated to be effective in text classification, sentiment analysis, and hate speech detection, as seen in the studies by Alatawi (2021), Alharbi (2021), and Saeed (2023). Additionally, word embedding has been applied to software development and bug prediction, with Wang (2024) and Devi (2025) utilizing deep learning models for bug assignment and software defect prediction. In bioinformatics, Zhenghui (2025) and Karakaya (2024) proposed models for bioactive peptide prediction, achieving high accuracy with ensemble learning and deep learning techniques. Furthermore, word embedding has been used in time series classification, with Tabassum (2022) proposing SAFE for time-series classification. In medical and health applications, Mostafa (2025) predicted drug-drug interaction seriousness using machine learning and semantic embedding. Overall, the studies in this cluster demonstrate the potential of word embedding in deep learning for various applications, highlighting its effectiveness in improving the accuracy and efficiency of tasks such as text classification, sentiment analysis, and bug prediction.

Subthemes:

• ***Natural Language Processing (NLP) Applications :*** (Rahman M. (2026) [50], Almuzaini H. (2020) [51], Alatawi H. (2021) [52], Alharbi N. (2021) [58], Saeed A. (2023) [59], Habib M. (2021) [61], Bayazed A. (2024) [62], Wei X. (2024) [63])
The applications of Natural Language Processing (NLP) are diverse and widespread, as seen in the provided citations. Rahman (2026) utilized word embedding feature extraction and CNN-LSTM to identify blood-brain barrier penetrating peptides. Almuzaini (2020) investigated the impact of stemming and word embedding on deep learning-based Arabic text categorization, finding that stem-based algorithms perform slightly better. Alatawi (2021) detected white supremacist hate speech using domain-specific word embedding and deep learning, achieving an F1-score of 0.79605 with BERT. Alharbi (2021) evaluated sentiment analysis via word embedding and RNN variants for Amazon online reviews, with the GLRNN algorithm and FastText feature extraction scoring the highest accuracy. Saeed (2023) used topic modeling and word embedding to detect Islamophobic textual content, with the proposed approach effectively detecting such content. Habib (2021) proposed an Arabic neural-based word embedding model for medical and health applications, with Word2Vec and fastText capturing the semantics of text more than GloVe. Bayazed (2024) introduced the ACOM approach for Arabic Comparative Opinion Mining, leveraging word embedding, deep learning models, and LLM-GPT, with BERT and BiLSTM achieving impressive performance. These studies demonstrate the potential of NLP in various applications, including text classification, sentiment analysis, and hate speech detection. Wei (2024) extracts event causality using external knowledge and polyhedral word embedding.

• ***Software Development and Bug Prediction :*** (Wang R. (2024) [53], Devi M. (2025) [54], Sepahvand R. (2023) [65])
Research in software development and bug prediction has focused on utilizing word embedding and deep learning models (Wang, 2024; Devi, 2025). These models have been applied to bug assignment (Wang, 2024) and software defect prediction (Devi, 2025). Additionally, convolution neural networks have been used for bug triaging, considering design flaws (Sepahvand, 2023). The use of bidirectional word embedding systems has also been explored (Devi, 2025). These studies demonstrate the potential of machine learning techniques in improving software development and bug prediction. The original title of the subtheme is: Software Development and Bug Prediction

• ***Bioinformatics and Peptide Prediction :*** (Zhenghui L. (2025) [55], Karakaya O. (2024) [60])
Zhenghui (2025) and Karakaya (2024) propose models for bioactive peptide prediction, achieving high accuracy with ensemble learning and deep learning techniques. Zhenghui (2025) integrates CNNs and BiGRUs, while Karakaya (2024) consolidates word embedding and BiLSTM, enhancing classification accuracy. Original title: Bioinformatics and Peptide Prediction

• ***Time Series Classification and Sentiment Analysis :*** (Tabassum N. (2022) [57])
Tabassum (2022) proposes SAFE for time-series classification.

• ***Medical and Health Applications :*** (Mostafa A. (2025) [64])
Mostafa (2025) predicts drug-drug interaction seriousness using machine learning and semantic embedding.

Theme 4 / Semantic Embedding for Multimodal and Complex Data

Summary:

This theme encompasses a wide range of research topics primarily focused on the application and development of semantic embedding techniques across various domains, including remote sensing, image-text retrieval, person re-identification, sarcasm detection, zero-shot learning, and emotion recognition in speech.

Synthesis of all articles in this group:

The cluster 'Semantic Embedding for Multimodal and Complex Data' encompasses a broad range of applications and techniques, as seen in the works of Wang (2021), Feng (2022), and Xie (2023). Remote sensing applications have benefited from enhanced feature pyramid networks for scene classification (Wang, 2021) and direction-oriented visual-semantic embedding models for image-text retrieval (Ma, 2024). Visual-semantic embedding approaches have been proposed for image-text retrieval, utilizing multimodal knowledge graphs (Feng, 2022) and attention mechanisms (Dong, 2023) to bridge the semantic gap between visual and textual data. Zero-shot learning has been explored in various domains, including natural images (Xie, 2023) and medical images (Paul, 2021), with methods such as balanced semantic embedding (Xie, 2023) and trait-guided multi-view semantic embedding (Paul, 2021) showing superiority over state-of-the-art counterparts. Additionally, multimodal and speech processing have been investigated, with hierarchical frameworks for multi-modal sarcasm detection (Fu, 2024) and zero-shot emotion recognition in speech (Xu, 2022) utilizing semantic information. Person re-identification and image classification have also been improved through visual-semantic embedding and parallel semantic analysis (Xiang, 2025; Yin, 2022). Overall, the cluster demonstrates the effectiveness of deep learning techniques and semantic embedding in various applications, as highlighted by Wang (2023), Qin (2024), and Liu (2022). The use of multimodal knowledge relations, attention mechanisms, and semantic gaps has been shown to enhance the embeddings of images and text, as noted by Feng (2022) and Dong (2023).

Subthemes:

• ***Remote Sensing Applications :*** (Wang X. (2021) [66], Ma Q. (2024) [67], Wang C. (2023) [81])

Remote sensing applications have seen significant advancements, with Wang (2021) proposing an enhanced feature pyramid network for scene classification. Ma (2024) introduced a direction-oriented visual-semantic embedding model for image-text retrieval, addressing visual-semantic imbalance. Wang (2023) developed a semantic embedding generative adversarial network for remote sensing image synthesis, improving visual fidelity and diversity. These methods demonstrate the effectiveness of deep learning techniques in remote sensing, as shown by their state-of-the-art results on various benchmarks.

• ***Visual-Semantic Embedding for Image-Text Retrieval :*** (Feng D. (2022) [69], Qin X. (2024) [79], Liu Y. (2022) [73], Liu Y. (2024) [80], Dong S. (2023) [78])
The citations discuss various approaches to visual-semantic embedding for image-text retrieval. Feng (2022) proposes a multimodal knowledge enhanced visual-semantic embedding approach, which utilizes implicit multimodal knowledge relations between images and text. Qin (2024) presents a multi-task visual semantic embedding network for image-text retrieval. Liu (2022) regularizes visual semantic embedding with contrastive learning for image-text matching. Liu (2024) focuses on prior semantic-embedding representation learning for on-the-fly fine-grained sketch-based image retrieval. Dong (2023) proposes a semantic embedding guided attention with explicit visual feature fusion for video captioning. These approaches aim to bridge the semantic gap between visual and textual data, as noted by Feng (2022) and Dong (2023). The use of multimodal knowledge graphs, as seen in Feng (2022), and attention mechanisms, as in Dong (2023), are effective in enhancing the embeddings of images and text. Overall, the citations demonstrate the importance of considering implicit multimodal knowledge relations and semantic gaps in visual-semantic embedding for image-text retrieval tasks, as highlighted by Qin (2024) and Liu (2022).

• ***Zero-Shot Learning and Generalization :*** (Gan Y. (2025) [68], Xie G. (2023) [74], Paul A. (2021) [76])
Zero-shot learning (ZSL) has been explored in various domains, including natural images (Xie, 2023) and medical images (Paul, 2021). ZSL models aim to synthesize unseen class features using only seen class feature and attribute pairs as training data (Xie, 2023). To address challenges in ZSL, researchers have proposed methods such as balanced semantic embedding (Xie, 2023) and trait-guided multi-view semantic embedding with self-training (Paul, 2021). Additionally, adversarial transferability has been boosted via latent encoding and semantic embedding perturbations (Gan, 2025). These approaches have shown superiority over state-of-the-art counterparts in various benchmarks.

• ***Multimodal and Speech Processing :*** (Fu H. (2024) [70], Xu X. (2022) [72])
Fu (2024) proposes a hierarchical framework for multi-modal sarcasm detection, while Xu (2022) explores zero-shot emotion recognition in speech. Both studies utilize semantic information, with Fu using sentiment dictionaries and Xu using semantic-embedding prototypes.

• ***Person Re-Identification and Image Classification :*** (Xiang S. (2025) [71], Yin P. (2022) [75])
Xiang (2025) and Yin (2022) propose methods for person Re-Identification and place recognition, utilizing visual-semantic embedding and parallel semantic analysis, respectively, to improve generalization capability and accuracy in various applications, including public security and autonomous driving, with Xiang (2025) introducing a dynamic masking mechanism and Yin (2022) using spherical harmonics domain.

Theme 5 / Word Embedding for Enhanced NLP Tasks

Summary:

This theme comprises citations that primarily focus on the application of word embedding techniques in various natural language processing (NLP) tasks. These tasks include text document summarization, news text classification, sentiment analysis, document clustering, patent recommendation, extractive text summarization, knowledge representation learning, legal judgment recommendation, text similarity measurement, and intrusion detection in telecommunication networks.

Synthesis of all articles in this group:

The cluster of Word Embedding for Enhanced NLP Tasks encompasses various subthemes, including Text Summarization and Similarity, Sentiment Analysis and Opinion Mining, Document Classification and Clustering, Knowledge Representation and Recommendation, and Network Security and Telecommunication. According to Zhang (2025), word embedding techniques can be used for daily tourism demand forecasting, while Mohd (2020) and Rani (2021) demonstrate the effectiveness of word embedding in text document summarization and extractive text summarization, respectively. Kumar (2022) combines Plutchik wheel and word embedding for Twitter sentiment analysis, highlighting the potential of word embedding in sentiment analysis. In document classification and clustering, Zhao (2022) proposes WTL-CNN for news text classification, combining word2vec and TF-IDF, and achieves a precision rate of 95.76%. Additionally, Chen (2020) and Wei (2022) explore word embedding and joint semantic embedding for knowledge representation and recommendation, while Yan (2025) proposes a Transformer-based intrusion detection system for post-5G and 6G networks, enhancing network security. Overall, the studies demonstrate the effectiveness of word embedding techniques in improving various NLP tasks, including text summarization, sentiment analysis, document classification, knowledge representation, and network security, as seen in the works of Yulianti (2023), Bai (2021), and Dhanani (2022). The use of weighted word embedding, such as in the Re-LSTM model proposed by Zhao (2022), can further enhance the performance of text summarization and similarity tasks. The applications of word embedding techniques are diverse, ranging from text analysis to network security, and have shown promising results in various domains.

Subthemes:

• ***Text Summarization and Similarity :*** (Zhang C. (2025) [82], Mohd M. (2020) [83], Farouk M. (2020) [85], Rani R. (2021) [89], Zhao W. (2022) [92], Yulianti E. (2023) [6])
The subtheme of Text Summarization and Similarity has been explored in various studies, including the use of word embedding techniques for daily tourism demand forecasting (Zhang, 2025) and text document summarization (Mohd, 2020). Measuring text similarity based on structure and word embedding has also been investigated (Farouk, 2020). Weighted word embedding has been used to improve extractive text summarization (Rani, 2021) and text similarity calculation (Zhao, 2022). The use of weighted word embedding has been shown to enhance the performance of text summarization algorithms, such as TextRank (Yulianti, 2023). The Re-LSTM model, which combines weighted word embedding with a long short-term memory network, has been proposed for text similarity computation (Zhao, 2022). Overall, the studies demonstrate the effectiveness of word embedding techniques in improving text summarization and similarity tasks. The original title of the subtheme is: Text Summarization and Similarity

• ***Sentiment Analysis and Opinion Mining :*** (Kumar P. (2022) [86])
Kumar (2022) combines Plutchik wheel and word embedding for Twitter sentiment analysis

• ***Document Classification and Clustering :*** (Zhao W. (2022) [84], Bai R. (2021) [87])
Zhao (2022) proposes WTL-CNN for news text classification, combining word2vec and TF-IDF, while Bai (2021) explores deep multi-view document clustering with enhanced semantic embedding, both aiming to improve document classification and clustering accuracy, as seen in Zhao's (2022) precision rate of 95.76% and Bai's (2021) clustering approach.

• ***Knowledge Representation and Recommendation :*** (Chen J. (2020) [88], Wei X. (2022) [90], Dhanani J. (2022) [91])
Research in knowledge representation and recommendation has explored various techniques, including word embedding (Chen, 2020) and joint semantic embedding (Wei, 2022). These methods aim to improve patent recommendation and knowledge representation learning. Additionally, pre-learned word embedding has been used for legal judgment recommendation, demonstrating effectiveness and scalability (Dhanani, 2022). These studies highlight the importance of incorporating domain-specific knowledge and distributed learning approaches to enhance recommendation systems.

• ***Network Security and Telecommunication :*** (Yan H. (2025) [93], Du L. (2023) [94])
Yan (2025) proposes a Transformer-based intrusion detection system for post-5G and 6G networks, while Du (2023) introduces DBWE-Corbat for background network traffic generation, both enhancing network security, as seen in the subtheme: Network Security and Telecommunication

Theme 6 / Specialized Applications of Word Embedding

Summary:

This theme consists of 8 citations that primarily focus on the application of word embedding techniques in various domains, including industrial alarm floods, research topic evolution, and natural language processing tasks such as duplicate question detection and bilingual lexicon induction.

Synthesis of all articles in this group:

The cluster of Specialized Applications of Word Embedding encompasses various subthemes, including Industrial Alarm Floods, Word Embedding Techniques, and Natural Language Processing. According to Hu (2024), word embedding can be used to address industrial alarm floods through root cause identification. Additionally, word embedding techniques have been explored for dimensionality reduction (Treistman, 2022) and bilingual lexicon induction (Cao, 2023). In the realm of Natural Language Processing, semantic embeddings have been utilized for duplicate question detection (Altamimi, 2024) and scene graph generation (Zhao, 2025). These studies demonstrate the versatility and advancements of word embedding in various applications, including industrial alarm management (Davoodi, 2025) and research topic analysis (Huang, 2024). Overall, the cluster highlights the potential of word embedding in specialized applications, enabling improved results and efficiency in diverse fields.

Subthemes:

• ***Industrial Alarm Floods :*** (Hu W. (2024) [95], Hu W. (2023) [97], Davoodi A. (2025) [99])
Industrial alarm floods can be addressed through root cause identification using word embedding and few-shot learning (Hu, 2024). Pattern matching can also be applied

using word embedding and dynamic time warping (Hu, 2023). A three-stage computational procedure can group and visualize correlated alarms, enabling operators to focus on root-cause abnormalities (Davoodi, 2025). This procedure includes time-augmented word embedding, density-based clustering, and hierarchical visualization. Original title of the subtheme: Industrial Alarm Floods

• ***Word Embedding Techniques :*** (Treistman A. (2022) [98], Huang S. (2024) [96], Cao H. (2023) [101])
Word embedding techniques have been explored by Treistman (2022) for dimensionality reduction using dynamic variance thresholding. Huang (2024) examined evolutions of semantic consistency in research topics via contextualized word embedding. Cao (2023) proposed bilingual word embedding fusion for robust unsupervised bilingual lexicon induction, achieving competitive performance on distant languages. These studies demonstrate advancements in word embedding techniques, as seen in Treistman (2022), Huang (2024), and Cao (2023).

• ***Natural Language Processing :*** (Altamimi A. (2024) [100], Zhao M. (2025) [102])
Altamimi (2024) uses Siamese MaLSTM and ELMO for duplicate question detection, achieving 95.68% accuracy. Zhao (2025) proposes a panoptic segmentation-based model for scene graph generation. Both studies utilize semantic embeddings for improved results.

Theme 7 / Word Embedding for Fake News Detection

Summary:

This theme focuses on fake news detection using word embedding techniques. The citations explore various approaches, including machine learning, deep learning, and linguistic feature-based methods.

Synthesis of all articles in this group:

The cluster explores the application of word embedding techniques for fake news detection, with Verma (2021) and Al-Tarawneh (2024) highlighting the use of TF-IDF, Word2Vec, and FastText. Linguistic features also aid in fake news detection, as noted by Sharma (2023). Furthermore, deep learning approaches, such as the Multi-Kernel Optimized Convolutional Neural Network proposed by Zaheer (2023), achieve high accuracy in detecting fake news. The combination of word embedding techniques and deep learning models, as seen in Al-Tarawneh (2024) and Zaheer (2023), demonstrates the potential for improved fake news detection. Overall, the cluster suggests that word embedding and deep learning techniques can be effective in identifying fake news, with TF-IDF and CNNs achieving high performance, as reported by Al-Tarawneh (2024).

Subthemes:

• ***Word Embedding for Fake News Detection :*** (Verma P. (2021) [103], Al-Tarawneh M. (2024) [104])
Word embedding techniques, such as TF-IDF, Word2Vec, and FastText, are used for fake news detection (Verma, 2021) (Al-Tarawneh, 2024). TF-IDF embeddings achieve high performance with SVMs and CNNs, while Word2Vec and FastText introduce complexities (Al-Tarawneh, 2024).

• ***Linguistic Feature-Based Word Embedding :*** (Sharma R. (2023) [105])
Linguistic features aid fake news detection (Sharma, 2023)

• ***Deep Learning for Fake News Detection :*** (Zaheer K. (2023) [106])
Zaheer (2023) proposes a Multi-Kernel Optimized Convolutional Neural Network for fake news detection with 85.8% accuracy

Miscellaneous / Articles that do not have any bibliometric communality with other texts in the map:

• (Ellaky Z. (2024) [1], Chen D. (2023) [2], Yang F. (2025) [4], Hasanzadeh S. (2020) [5], Venu Gopalachari M. (2023) [7], Rao R. (2021) [8])

Summary:

These articles could have some communalities with previously identified themes but communalities could not be assessed through bibliometric analyses (ARTIREV). Hence, communalities are solely assessed with the help of agentic AI (SOCRATES), who will propose some classification that will have to be verified by the user

Synthesis of all articles in this group:

The citations provided showcase a range of applications for word embedding techniques in various fields, including social media bot detection (Ellaky, 2024), IoT intrusion detection (Chen, 2023), cross-media hashing (Yang, 2025), review-based recommender systems (Hasanzadeh, 2020), aspect-based sentiment analysis (Venu Gopalachari, 2023), and phishing website detection (Rao, 2021). According to Ellaky (2024), the use of word embedding and deep learning architectures can achieve high accuracy in detecting social media bots. Similarly, Chen (2023) demonstrates the effectiveness of fusion word embedding deep transfer learning in heterogeneous IoT intrusion detection. The application of word embedding in review-based recommender systems, as proposed by Hasanzadeh (2020), can improve the accuracy of rating predictions. Furthermore, Venu Gopalachari (2023) highlights the importance of word embedding in aspect-based sentiment analysis for multi-domain reviews. Rao (2021) also utilizes word embedding and machine learning to detect phishing websites. Overall, these studies demonstrate the versatility and potential of word embedding techniques in addressing various challenges in the field of engineering sciences, as evidenced by the works of Yang (2025) and others.

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